

Hold, Breathe, Release - Exploring Tangible Interfaces for Mindful Breaks in Remote Work Environments

Katarzyna Wojna-Nowak
Institute of Applied Computer Science
University of Technology
Lodz, Poland
katarzyna.wojna-nowak@p.lodz.pl

Anna Kubczak
Institute of Applied Computer Science
University of Technology
Lodz, Poland
anna.kubczak@p.lodz.pl

Julia Dominiak
University of Technology
Lodz, Poland
julia.dominiak@p.lodz.pl



Figure 1: The teaser illustrates the early stage of the *Breathing Stone* prototype: a 3D - printed tangible artifact connected to an Arduino-based system. Once activated, it emits a gentle pulsing light—simulating a slow breath — and invites users to synchronize their rhythm with its calm, multisensory cues.

Abstract

Digital well-being technologies for remote work often frame rest as a resource to be measured and optimized. While tangible and affective interaction research has proposed embodied alternatives, less attention has been paid to how materiality and programmability shape experiences of calm and care. This paper presents an early design inquiry through the *Breathing Stone*, a tangible artifact used as a design probe rather than a finished system. The study

surfaces tensions around material choices, authenticity, and familiarity in calm technologies, questioning dominant assumptions in well-being design and outlining open questions for future material explorations.

CCS Concepts

• Human-centered computing → Interaction design theory, concepts and paradigms.

Keywords

Tangible artifact, Mental - health well-being, 3D prototype, Concept, Early Design



This work is licensed under a Creative Commons Attribution 4.0 International License.
CHI EA '26, Barcelona, Spain

© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2281-3/26/04
<https://doi.org/10.1145/3772363.3798947>

ACM Reference Format:

Katarzyna Wojna-Nowak, Anna Kubczak, and Julia Dominiak. 2026. Hold, Breathe, Release - Exploring Tangible Interfaces for Mindful Breaks in Remote Work Environments. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3798947>

1 Introduction and Background

The global shift toward remote work has transformed the rhythms of knowledge work, blurring the boundaries between focus and fatigue, between productivity and presence [30]. As screens increasingly mediate our cognitive and social worlds, concerns over digital well-being have intensified [18]. A growing ecosystem of well-being tools now promises to mitigate sensory overload and mental exhaustion, yet their impact remains uneven and often fleeting [24, 26]. While mindfulness and meditation apps have demonstrated modest gains in psychological flexibility and awareness, recent meta-analyses urge a move beyond purely screen-based interventions, highlighting their limited engagement and disembodied nature [7, 11, 17]. Amid this landscape, the concept of *mindful micro-breaks* has emerged as a small but potent countermeasure [2, 23]. These brief, intentional pauses - sometimes no longer than ten minutes - have been shown to restore energy, enhance focus, and rebalance emotion throughout the workday. Yet their practice remains constrained by workplace cultures that valorize constant busyness, where even resting feels like a transgression. This tension opens a design opportunity: how might we invite restorative pauses that integrate seamlessly into the flow of digital work, without demanding explicit attention or time commitment [22, 28]?

Within this context, tangible interaction offers a compelling alternative. By reintroducing materiality and embodied engagement into digital environments, tangible artifacts can transform well-being from an abstract metric into a sensory experience [12]. Recent explorations - from biophilic homeware to interactive objects that respond to touch and light - demonstrate how physicality and ambient feedback can anchor presence amid the abstraction of remote work [9, 12, 16, 33]. Unlike prior tangible well-being systems that categorize or evaluate interaction strategies (e.g., Claisse et al., 2022; Gennari et al., 2023), this work uses a single artifact to interrogate tensions between material authenticity and programmability. Among these, projects such as *Manifesting Breath* [13] demonstrate how shape-changing biofeedback artifacts can externalize physiological processes like breathing to support emotional regulation. Building on this line of inquiry, our work extends the notion of breath-based interaction into a speculative, ambient form designed for everyday remote work contexts. Moreover, research in this domain has highlighted how embodied interaction enables more inclusive and valid ways of assessing attention and cognitive load, enriching the landscape of well-being technologies beyond mere screen interaction [20].

However, despite this growing body of work, there remains a limited understanding of how **material properties** (e.g., natural versus synthetic) and degrees of programmability shape the perceived sincerity, comfort, and longevity of calm technologies—particularly in everyday, non-therapeutic work contexts. Much of the literature

implicitly associates calm with natural materials and aesthetic authenticity, while simultaneously relying on programmable systems to enable interaction and responsiveness. This unresolved tension has received little direct attention. This paper addresses these questions through an ongoing design inquiry centered on the *Breathing Stone*, an early tangible artifact intended to invite brief moments of pause during remote work. Importantly, we do not present the *Breathing Stone* as a finished design or a novel interaction technique. Instead, we use it as a vehicle to articulate emerging design tensions and to reframe calm technologies as sites of inquiry rather than solutions.

2 Concept and Design Inquiry

This work adopts a design-as-inquiry perspective, drawing on slow technology, calm computing, and speculative design. Rather than validating outcomes or optimizing behavior, the focus is on surfacing values, assumptions, and trade-offs in material and interaction decisions. Artifacts are treated as epistemic tools that materialize questions and make experiential tensions visible. The current prototype includes a 3D-printed shell, an Arduino microcontroller, and an LED module enabling programmable breathing-like light patterns. Vibration has been technically tested but is not yet integrated in the casing. Thermal feedback remains conceptual and will be explored in future iterations.

The *Breathing Stone* is approached not as a finalized interface, but as an exploratory design probe investigating how calm and care might be sensed or interpreted through tangible, ambient interaction in everyday remote work. Its current status is tentative; contributions lie in problem framing and articulating open design questions rather than empirical claims.

The Breathing Stone as a Design Probe. The *Breathing Stone* examines how calm is negotiated at the intersection of materiality, programmability, and work rhythms. Minimal, non-instrumental cues invite rather than command attention, shifting focus from measurable impact to perceived qualities such as rhythm, warmth, and responsiveness, shaped by material expectations and cultural associations.

Form and Material Explorations. Currently, the artifact is a smooth, pebble-like object that fits in the hand, evoking natural associations with calm and stability. Fabricated from bioplastic or soft silicone, it supports rapid iteration and actuation. This reflects a central tension: natural materials convey authenticity but resist programmability, while synthetic materials allow responsiveness but risk perceived artificiality. The *Breathing Stone* intentionally occupies this ambiguous space to explore the interplay of authenticity and programmability.

Interaction and Temporal Behaviour. Interaction emphasizes rhythm and warmth over textual or auditory feedback. The artifact produces a slow, cyclical vibration and warmth that loosely suggests breathing, leaving interpretation open. Hand contact subtly modulates feedback, enabling co-regulation without measurement. Interactions are brief (one minute) and taper gradually, framing the experience as embedded micro-pauses rather than tasks. Its internal logic remains conceptual (idle, invitation, engagement, closure) as a scaffold for exploring rhythm and transition.

Speculative Use Vignettes. **Between Meetings.** After video calls, the user touches the *Breathing Stone*, experiencing a brief, rhythmic pause before it returns to idle.

During Overstimulation. Rising digital noise prompts subtle changes in rhythm, noticed peripherally, suggesting a pause without requiring action.

End-of-Day Transition. The artifact slows and dims as the workday ends, marking a soft boundary between work and personal time.

Across these scenarios, the *Breathing Stone* functions as an ambient presence supporting transitions, emphasizing attunement and presence over productivity (see Figure 2).



The image presents the first scenario, where the user touches the *Breathing Stone*.

(b) The image presents the second scenario, where the Stone subtly “inhales.”

(c) The image presents the third scenario, where the Stone slows its rhythm.

Figure 2: The collage of three images presents three scenarios.

The design of the *Breathing Stone* draws on the principles of *slow technology* [3, 21] and *calm computing* [8, 29], both of which advocate for technologies that foster reflection, presence, and temporal depth. These paradigms challenge the dominant logic of acceleration by proposing artifacts that engage users in contemplation rather than continuous productivity. Within the contemporary landscape of remote work—characterized by relentless screen time, fragmented attention, and ambient digital noise—the *Breathing Stone* operates as a counterpoint. It seeks not to optimize time, but to expand it; not to capture attention, but to return it to the body.

The artifact is conceived as a mediator of pause through multisensory subtlety. Its slow rhythmic vibration, gentle warmth, and dim illumination are calibrated to exist at the edge of perception—barely noticeable, yet emotionally legible. This deliberate minimalism resists the overt interactivity of conventional well-being technologies. Instead of issuing commands or feedback loops, the Stone communicates through affective texture and rhythm, inviting a quiet synchronization between the object and its holder. In this sense, interaction becomes a form of co-regulation, where both artifact and user share a breathing space rather than a task. This aligns with recent explorations in tangible interaction that frame physical artifacts as companions in affective attunement and self-reflection rather than as control interfaces [10, 31, 32].

The physical form further reinforces this ethos. The Stone’s organic shape and smooth tactility evoke the calming familiarity of natural materials—river pebbles, weathered rocks, or ceramics warmed by touch. Such biomimetic softness stands in quiet defiance

of the angular, metallic aesthetics of most digital devices. This material gesture is intentional: it communicates care and temporal slowness, suggesting that technology, too, can embody qualities of tenderness and rest. The *Breathing Stone* thus becomes both a symbolic and sensory interruption in the rhythm of digital labor—a designed invitation to slow down, hold, and simply breathe.

Ultimately, the rationale behind this design is not merely functional but philosophical. It asks whether technologies might act not as accelerators of productivity, but as companions in restoration. In a culture where attention is perpetually externalized, the *Breathing Stone* proposes a return to the body as an interface of awareness. It transforms interaction from an act of control into an act of care.

3 Discussion: Toward Open Research Questions

This Work in Progress situates the *Breathing Stone* within ongoing discourses on calm, embodiment, and materiality in Human–Computer Interaction. Rather than offering conclusions about effectiveness or user outcomes, the project uses an early-stage artifact to surface tensions that arise when designing for calm in digitally saturated work environments. In doing so, it treats calm not as a state to be produced, but as a fragile experiential quality shaped by material, temporal, and cultural conditions. Across prior research, calm has been framed as a desirable yet elusive design quality. Calm computing and slow technology emphasize backgrounded interaction, temporal depth, and reflective engagement as alternatives to attention-grabbing systems [8, 19, 29]. Similarly, work on embodied and tangible interaction highlights how physical artifacts can support meaning-making through sensory engagement rather than explicit instruction [25, 31]. The *Breathing Stone* resonates with these perspectives while also exposing their unresolved questions—particularly around how calm is recognized, trusted, and sustained in everyday use.

Programmability and the Ambiguity of Care. One of the central tensions surfaced by this inquiry concerns programmability. Prior work on shape-changing interfaces and biofeedback artifacts suggests that dynamic responsiveness can enhance bodily awareness and emotional regulation [13]. However, these systems often rely on explicit sensing, feedback, and goal-oriented interaction, which may foreground instrumentation and measurement. The *Breathing Stone* complicates this lineage by employing programmability in a deliberately restrained and ambiguous manner. Its rhythms and thermal shifts do not correspond directly to measured physiological states, nor do they promise improvement or optimization. This raises an open question: when does programmability feel supportive, and when does it begin to undermine perceived care by revealing the computational logic beneath the interaction? The artifact suggests that programmability may be most compatible with calm when it remains partially opaque—felt rather than understood—yet this opacity also risks disengagement or misinterpretation.

Material Expectations and Trust. Materiality emerged as another key site of reflection. Research on product experience and attachment has long emphasized that material qualities shape emotional response, trust, and perceived value [1, 14]. In the context of well-being technologies, natural materials are frequently associated with

authenticity, calm, and care, while synthetic materials enable actuation and adaptability. By adopting a soft, pebble-like form fabricated from synthetic materials, the *Breathing Stone* intentionally occupies an ambiguous material space. This choice foregrounds a tension between aesthetic familiarity and technical affordance. Rather than resolving this tension, the artifact renders it experientially available, prompting reflection on whether calm must be materially “honest” to be felt as sincere. This raises further questions about how users negotiate trust when computational behavior is embedded in forms that visually reference natural or contemplative objects.

Scale, Duration, and the Everyday. Unlike many well-being interventions that frame calm as a dedicated activity—such as meditation sessions or guided breathing exercises—the *Breathing Stone* explores calm at the scale of brief, incidental encounters. Its interactions are intentionally short, lasting approximately one minute, and do not require goal setting, progress tracking, or explicit intention. This micro-temporal framing aligns with research on everyday rituals and peripheral interaction, which suggests that meaning can emerge through repetition and familiarity rather than intensity [5, 27]. However, it also raises questions about visibility and relevance: how subtle can an interaction become before it fades into irrelevance? And how might such artifacts be appropriated over time—ritualized, ignored, or repurposed?

Resisting Productivity-Oriented Framings of Rest. The *Breathing Stone* also intersects with critical perspectives on digital well-being. Many contemporary tools frame rest instrumentally, positioning it as a means to improve focus, resilience, or productivity [4, 6]. While such framings may increase adoption, they risk reinforcing the same logics of optimization that contribute to exhaustion. This project explores whether intentional limitation—such as the absence of metrics, goals, or feedback—might function as a design value in itself. By offering pause without justification, the artifact raises questions about refusal and restraint as legitimate design strategies. Rather than optimizing rest, it probes how rest might be acknowledged as valuable without being made productive.

Toward a Problem Space for Tangible Calm. Taken together, these reflections suggest that designing for calm involves navigating unresolved tensions rather than converging on stable solutions. The *Breathing Stone* does not resolve questions of material authenticity, programmability, or effectiveness. Instead, it surfaces them as interdependent concerns that shape how calm is experienced and interpreted in everyday work contexts. This Work in Progress therefore contributes a problem space for tangible well-being research, articulated through an exploratory artifact. The questions raised extend beyond this specific design and invite further investigation:

- When does programmability enhance experiences of calm, and when does it undermine perceived care or authenticity?
- How do material expectations influence trust and long-term engagement with tangible well-being technologies?
- How small, brief, or peripheral can an interaction be while still supporting meaningful experiences of pause?
- What roles can limitation, ambiguity, and refusal play in resisting productivity-driven framings of rest?

Rather than offering answers, this work positions calm as a fragile, negotiated experience—one shaped by material, temporal, and cultural design decisions that warrant continued exploration.

3.1 Next Steps

Future work will focus on systematic material explorations, including comparative prototyping with natural materials such as stone and wood, as well as hybrid forms combining natural shells with programmable cores. In parallel, exploratory field engagements in home-office settings will examine how such artifacts are appropriated over time. Rather than seeking immediate user validation of effectiveness, these steps aim to deepen understanding of how calm is perceived, negotiated, and sustained in everyday work contexts.

3.2 Limitations and Scope

This Work in Progress is framed as a conceptual, exploratory design inquiry. The *Breathing Stone* exists in an early stage of material and interaction exploration, with no user studies or long-term deployments conducted. Accordingly, this work does not aim to produce empirical claims about user experience or well-being outcomes. Following Gaver’s *design as inquiry*, conceptual artifacts can function as probes that surface values, assumptions, and tensions in everyday technologies [15]. Here, the *Breathing Stone* is used to explore how calm, care, and pause might be materially articulated, rather than to validate their effects. This approach entails trade-offs: experiential qualities such as comfort, subtlety, and engagement require situated, lived interaction over time. Future work will involve iterative prototyping and exploratory user engagement to examine how these speculative qualities are perceived, negotiated, or resisted, extending the problem framing presented here.

4 Conclusion

This paper presented an ongoing design inquiry into tangible well-being artifacts, using the *Breathing Stone* as a probe to surface unresolved tensions around materiality, programmability, and care. By reframing calm technologies as sites of inquiry rather than solutions, the work invites reflection on how HCI might design for presence without instrumentalization. As a Work in Progress, it offers not answers, but carefully situated questions intended to guide future exploration within CHI and beyond.

Disclaimer: ChatGPT-5.2 was used to create Figure 2.

References

- [1] Sahar Ahmadian, Bashir Sahraei, and Shekoufeh Khojasteh Khosro. 2023. Brand attachment, brand experience, brand image, perceived quality, perceived value, and brand loyalty. *Journal of Humanities Social Sciences And Business (JHSSB)* 3, 1 (2023), 257–269.
- [2] Patricia Albulescu, Irina Macsinga, Andrei Rusu, Coralia Sulea, Alexandra Bodnaru, and Bogdan Tudor Tulbure. 2022. "Give me a break!" A systematic review and meta-analysis on the efficacy of micro-breaks for increasing well-being and performance. *PLoS one* 17, 8 (2022), e0272460.
- [3] Armaghan Behzad Behbahani. 2019. *Designing for Reflection: Utilizing slow technology to create tangible interactive designs for reducing technostress*. Ph.D. Dissertation. Virginia Tech.
- [4] Timebot Blog. 2026. *How Digital Tools Are Redefining Workplace Productivity*. <https://blog.timebot.chat/how-digital-tools-are-redefining-workplace-productivity>. Accessed: 2026-01-22.
- [5] J Bostrand, S Ehrling, N Jasmani, R Sato, and F Sjöberg. [n. d.]. The Mere Exposure Effect on University Students. ([n. d.]).

- [6] Ph.D. Bryan Robinson. 2025. 'Digital Tool Fatigue,' *Eroding Mental Health and Career Productivity*. <https://www.forbes.com/sites/bryanrobinson/2025/10/04/digital-tool-fatigue-eroding-mental-health-and-career-productivity/> Accessed: 2026-01-22.
- [7] Ana I. Canhoto, Danae Manika, Emma Wainwright, and Donna-Marie Holder. 2025. The impact of technology: how features, resources and task demands shape digital well-being. *Information Technology & People* (2025). doi:10.1108/ITP-01-2024-0077 Ahead of print.
- [8] Amber Case. 2021. Calm Technology/Media and the Limits of Attention. In *MEDIA*. Intellect, 165–172.
- [9] Caroline Claisse, Muhammad Umair, Abigail C Durrant, Charles Windlin, Pavel Karpashevich, Kristina Höök, Vasiliki Tsaknaki, Pedro Sanches, and Corina Sas. 2022. Tangible interaction for supporting well-being. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts*. 1–5.
- [10] Cammy Crolic and Chris Janiszewski. 2023. Cognitive and affective reflection increases appreciation for less preferred subcategories of experiential goods. *Frontiers in Psychology* 14 (2023), 1271516.
- [11] Jun-jie Dong, Zhi-dan Tan, Yu-liang Zhang, Yu-jie Sun, and Yuan-kai Huang. 2025. Work from home and employee well-being: a double-edged sword. *BMC psychology* 13, 1 (2025), 748.
- [12] Daniel Echeverri and Huaxin Wei. 2023. Exploring the experience with tangible interactive narrative: Authoring and evaluation of Letters to José. *Entertainment Computing* 44 (2023), 100535.
- [13] Alexz Farrall, Jordan Taylor, Ben Ainsworth, and Jason Alexander. 2023. Manifesting breath: Empirical evidence for the integration of shape-changing biofeedback-based artefacts within digital mental health interventions. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–14.
- [14] Salma Fitri and Dean Salomo Anthonino Kumenaung. 2025. Brand attachment in mediating of customer experience, hedonic value, and personalized experience on customer retention. *Jurnal STEI Ekonomi* 34, 2 (2025), 179–192.
- [15] Bill Gaver, Tony Dunne, and Elena Pacenti. 1999. Design: cultural probes. *interactions* 6, 1 (1999), 21–29.
- [16] Rosella Gennari, Maristella Matera, Diego Morra, Alessandra Melonio, and Mehdi Rizvi. 2023. Design for social digital well-being with young generations: Engage them and make them reflect. *International Journal of Human-Computer Studies* 173 (2023), 103006.
- [17] Emanuela Ingusci, Fulvio Signore, Claudio Giovanni Cortese, Monica Molino, Paola Pasca, and Enrico Ciavolino. 2022. Development and validation of the remote working benefits & disadvantages scale. *Quality & Quantity* 57, 2 (2022), 1159.
- [18] Thomas Jacquet, Romuald Lepers, Benjamin Pageaux, and Bénédicte Poulin-Charronnat. 2023. Acute smartphone use impairs vigilance and inhibition capacities. *Scientific reports* 13, 1 (2023), 23046.
- [19] Reinhard Mehring. 2022. From Attention-Grabbing to Calm Technology. *Morals & Machines* 1, 2 (2022), 70–77.
- [20] Cornell University Meta Design & Technology Lab. 2025. Tangible interaction – Tag page. <https://mdtl.human.cornell.edu/tag/tangible-interaction/>. Accessed: 2025-10-24.
- [21] William Odom, Erik Stolterman, and Amy Yo Sue Chen. 2022. Extending a theory of slow technology for design through artifact analysis. *Human-Computer Interaction* 37, 2 (2022), 150–179.
- [22] Sonja Olson. 2024. Mindful micro-breaks—The potential of a 10-minute "paws". *NEWStat (American Animal Hospital Association)* (23 May 2024). <https://www.aaha.org/newstat/publications/mindful-micro-breaks-the-potential-of-a-10-minute-paws/> Published online 23 May 2024.
- [23] Yvette Prior, F Martelli Michael, and Juanita Anderson. 2021. Coping with work stress by using micro-breaks and reframing strategies: A qualitative study among healthcare workers. In *American Psychological Association (APA) Conference. Division*, Vol. 14.
- [24] Kus Hanna Rahmi, Adi Fahrudin, Tugimin Supriyadi, Elin Herlina, Rosilawati Rosilawati, and Sri Ratna Ningrum. 2025. Technostress and cognitive fatigue: Reducing digital strain for improved employee well-being: A literature review. *Multidisciplinary Reviews* 8, 12 (2025), 2025380–2025380.
- [25] Lovise Søyland. 2021. Grasping materialities: Making sense through explorative touch interactions with materials and digital technologies. (2021).
- [26] T Supriyadi, S Sulistiasih, KH Rahmi, B Pramono, and A Fahrudin. 2025. The Impact Of Digital Fatigue On Employee Productivity And Well-Being: A Scoping Literature Review. *Environment And Social Psychology* 10, 2 (2025).
- [27] Nada Terzimehić, Renate Häußlschmid, Heinrich Hussmann, and MC Schraefel. 2019. A review & analysis of mindfulness research in HCI: Framing current lines of research and future opportunities. In *Proceedings of the 2019 CHI conference on human factors in computing systems*. 1–13.
- [28] Katie Waxman. 2023. The Power of Micro-Breaks at Work. *CoolBreakrooms Blog* (14 Dec 2023). <https://coolbreakrooms.com/the-power-of-micro-breaks-at-work/> Published online 14 December 2023.
- [29] Mark Weiser and John Seely Brown. 1997. The coming age of calm technology. In *Beyond calculation: The next fifty years of computing*. Springer, 75–85.
- [30] John Wells, Florian Scheibein, Leonor Pais, Nuno Rebelo dos Santos, C-Andreas Dalluege, Jan Philipp Czakert, and Rita Berger. 2023. A systematic review of the impact of remote working referenced to the concept of work–life flow on physical and psychological health. *Workplace health & safety* 71, 11 (2023), 507–521.
- [31] S. A. G. Wensveen. 2005. *A Tangibility Approach to Affective Interaction*. Ph.D. Dissertation. Delft University of Technology. <https://resolver.tudelft.nl/uuid:6c1677bd-048b-407c-ab64-3ed9bc329c24> Doctoral thesis.
- [32] Jifan Yang, Steven Bednarski, Alison Bullock, Robin Harrap, Zack MacDonald, Andrew Moore, and TC Nicholas Graham. 2024. Fostering the AR illusion: a study of how people interact with a shared artifact in collocated augmented reality. *Frontiers in Virtual Reality* 5 (2024), 1428765.
- [33] Nianmei Zhou, Steven Devleminck, and Luc Geurts. 2024. Tangible affect: A literature review of tangible interactive systems addressing human core affect, emotions and moods. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference*. 424–440.